

RUTIKA AVINASH KADAM

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EDUCATION

Stony Brook University

Master of Science in Data Science

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, Agentic AI, Big Data.

Savitribai Phule Pune University

Bachelor of Engineering in Information Technology

Coursework: Discrete Structures, Data Structures & Algorithms, System Design, Cloud Computing.

Stony Brook, USA

August 2024 – May 2026

Pune, India

August 2016 – May 2020

SKILLS

Programming Languages: Python, R, SAS, SQL (MySQL, PostgreSQL, PL/SQL), C++, HTML, CSS

Data Science: NumPy, Pandas, Seaborn, scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face, LangChain, LangGraph

Data Engineering & Big Data: Azure Data Factory, Data Lake, Databricks, Microsoft Fabric, SSMS, SSIS

Cloud & DevOps: Azure, FastAPI, REST APIs, Docker, Kubernetes, MLflow, Git, GitHub, Postman

Visualization & Analytics: Power BI, Tableau, Microsoft Excel

Security & IT Operations: Qualys VMDR, Microsoft Endpoint Configuration Manager (SCCM), ServiceNow, Jira, Aternity, Vulnerability Management, Patch Management

PROFESSIONAL EXPERIENCE

Stony Brook Medicine

Data Scientist (Graduate Researcher)

Stony Brook, USA

July 2025 – Present

- Built an end-to-end **Multi-Modal RAG pipeline** over **6k+ pages of medical documents** for Department of Public Health, Stony Brook Medicine, using **Azure Document Intelligence** for layout-aware extraction & **GPT-4o** for figure captioning.
- Implemented **custom semantic chunking** using embedding-based sentence similarity & adaptive breakpoint detection.
- Designed & benchmarked **FLAT, HNSW, & IVF_FLAT indexes** in Milvus across **27k+ multi-modal chunks**, comparing p50/p95 latency & Recall@5; achieved comparable accuracy with approximate indexes at **~2.5× lower latency** than FLAT.
- Evaluated **vector search, MMR reranking, & hybrid Vector+BM25 retrieval with RRF** on a synthetic ground-truth dataset spanning text, tables, & images; achieved **0.90 Recall@5** with hybrid retrieval.
- Built a **grounded RAG generation pipeline** with structured JSON outputs & source citations, achieving **RAGAS scores of 0.82 faithfulness, 0.82 answer relevancy, 0.93 context precision, & 0.74 context recall**.

Tata Consultancy Services

System Analyst

Pune, India

August 2020 – April 2024

- Collaborated with Vulnerability Management, performed **exploratory data analysis** on **1 Million+ system vulnerability records** reporting from **Qualys VMDR** using **Python & MySQL**, uncovering trends, anomalies, & threat vectors.
- Reduced **security risk exposure by 15%** by developing **Linear & Logistic Regression models** predicting exploitability & patch prioritization using engineered vulnerability features.
- Discovered recurring remediation patterns across **1M+ records** by applying **semantic clustering using E5 embeddings with HDBSCAN** on remediation text data.
- Fine-tuned **DistilBERT classifier** on cluster-derived labels to predict remediation categories in real time, enabling automated ticket routing, cutting **manual triage by 40%**.
- Improved vulnerability monitoring visibility by designing **Power BI dashboards with DAX-driven KPIs** including **CVSS score, exploitability risk, remediation timelines, & compliance metrics**.
- Achieved 99% patch compliance across enterprise systems by deploying automated remediation workflows through Microsoft Endpoint Configuration Manager & automated tasks using PowerShell, boosting productivity by 25%.

Zensar Technologies

Data Intern

Pune, India

May 2019 – July 2019

- Implemented **ETL pipelines in Azure Data Factory with incremental batch loading** to ingest **multi-channel website traffic & operational data** into **Azure Data Lake Gen2** for scalable data integration for **Swiggy-India**.
- Developed **Power BI dashboards** to visualize **conversion funnel metrics & marketing channel attribution**, identifying drivers of traffic variability & campaign performance, increasing **high-intent traffic by 15%**.

PROJECT EXPERIENCE

EV Battery Capacity Prediction | PyTorch, Data Version Control, MLflow, FastAPI, Docker, GitHub Actions, Azure

- Estimated **electric vehicle battery capacity** from ordinary charging data, since direct measurement needs controlled conditions that occur in only **56%** of real sessions; trained deep learning models in **PyTorch** on **111K** charging windows from 30 vehicles, using **5-fold cross-validation** split by vehicle.
- Compared **LSTM** and **Transformer** models against simple baselines across **10 tracked experiments**, finding a linear model on odometer reading alone performed best (**RMSE 1.10** vs 1.69 and 1.71) — with a **positional-encoding test** confirming the limit was the 21-minute data window rather than the model.
- Versioned data and models with **DVC** across two cloud remotes, tracked every experiment in **MLflow**, and deployed a **FastAPI** service via **Docker** and **Azure Container Registry** to **Azure Container Apps** on a public HTTPS endpoint, with **CI** running **43 tests** on each push.